

Warm-up 1

a) Compute the taxicab (L^1) norm of the following $\mathbf{v} \in \mathbb{C}^2$:

$$\mathbf{v} = \begin{pmatrix} i \\ 2 + i \end{pmatrix}$$

b) Compute the following:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} =$$

During class 1

The following pair of vectors spans a subspace of $\mathbb{R}^3$. Find an orthonormal basis for this subspace by carrying out the Gram–Schmidt process on this pair of vectors:

$$\begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$